

Run Report: 2026090511362948895a

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1. Configuration

parameter	value
numrobots	13
seed	42
stopmode	workload
targettasks	2000
maxticks	100000
scheduler	totalcost
pathplanner	bfs
conflictmanager	localyield
demandmode	legacy
layoutpreset	default
chargercapacity	4
failureenabled	False
fastmode	True

2. Performance KPIs

metric	value
stopreason	targetreached
simulationticks	5588
taskscreated	2123
tasksassigned	2262
taskscompleted	2000
tasksfailed	0
averagethroughput	0.358
averagewaittime	150.900
averagecycletime	175.572
averagerobotutilization	0.907
totalreplans	1087
totalblockedticks	4972
blockedtimeseconds	1491.600
chargingevents	270
chargerwaitticks	619
batterytaskinterruptions	252
failuretaskinterruptions	0
taskreassignments	245

3. Distributions

metric	count	mean	median	p90	p95	max
taskwaitingtime	2010	144.965	4.000	635.200	760.100	910.000
taskcycletime	2000	175.572	36.000	675.200	803.050	963.000
robotblockedepisodeticks	13	5507.923	5518.000	5553.000	5557.800	5565.000
systemactivefraction	5588	0.771	0.769	0.923	1.000	1.000
robotutilization	13	0.907	0.911	0.915	0.915	0.916

4. Spatial Bottleneck Analysis

Coverage: 13/13 blocked events (100.0%).

rank	cell	blockedevents	blockedticks
1	(5, 8)	2	10958
2	(6, 4)	1	5565
3	(19, 1)	1	5553
4	(10, 11)	1	5553
5	(13, 4)	1	5525

Blocked ticks are attributed to the cell where blocking started. Hotspots near chargers indicate capacity problems; hotspots at aisle intersections indicate traffic-control problems.